

M105

Estimaciones en la dispersión de muestras.

σ^2 : parámetro poblacional a estimar

S^2 : estadístico muestral (var. aleatoria)

s^2 : estimador puntual de σ^2 para una muestra de tamaño n (el valor que toma S^2 para una muestra en específico).

$$\chi^2 = \frac{(n-1)S^2}{\sigma^2}, \quad (i) \quad \swarrow n-1$$

χ^2 -chi cuadrada con $\nu = n-1$ g.l.

$$P(\chi_{1-\alpha/2}^2 < \chi^2 < \chi_{\alpha/2}^2) = 1-\alpha \quad (ii)$$



De (i) y (ii) se obtiene el intervalo de confianza para la var. poblacional:

$$\frac{(n-1)S^2}{\chi_{\alpha/2}^2} < \sigma^2 < \frac{(n-1)S^2}{\chi_{1-\alpha/2}^2}$$

$$zarray = n.p.\text{chisquare}(0, 10, 1000)$$

↓
0.000001

$$\int_0^{\chi_{1-\alpha/2}^2} h(x; \nu) dx = \alpha/2$$

$$\int_0^{\chi_{\alpha/2}^2} h(x; \nu) dx = 1 - \frac{\alpha}{2}$$

Ejemplo.- Los siguientes pesos, en decagramos, de 10 paquetes de semillar para césped de cierta marca son: 46.4, 46.1, 45.8, 47.0, 46.1, 45.9, 45.8, 46.9, 45.2 y 46.0. Encuentre el intervalo de confianza del 95% para la varianza de todos los paquetes de césped que distribuye la marca. Explique por qué es importante que los pesos se ajusten con una dist. normal.

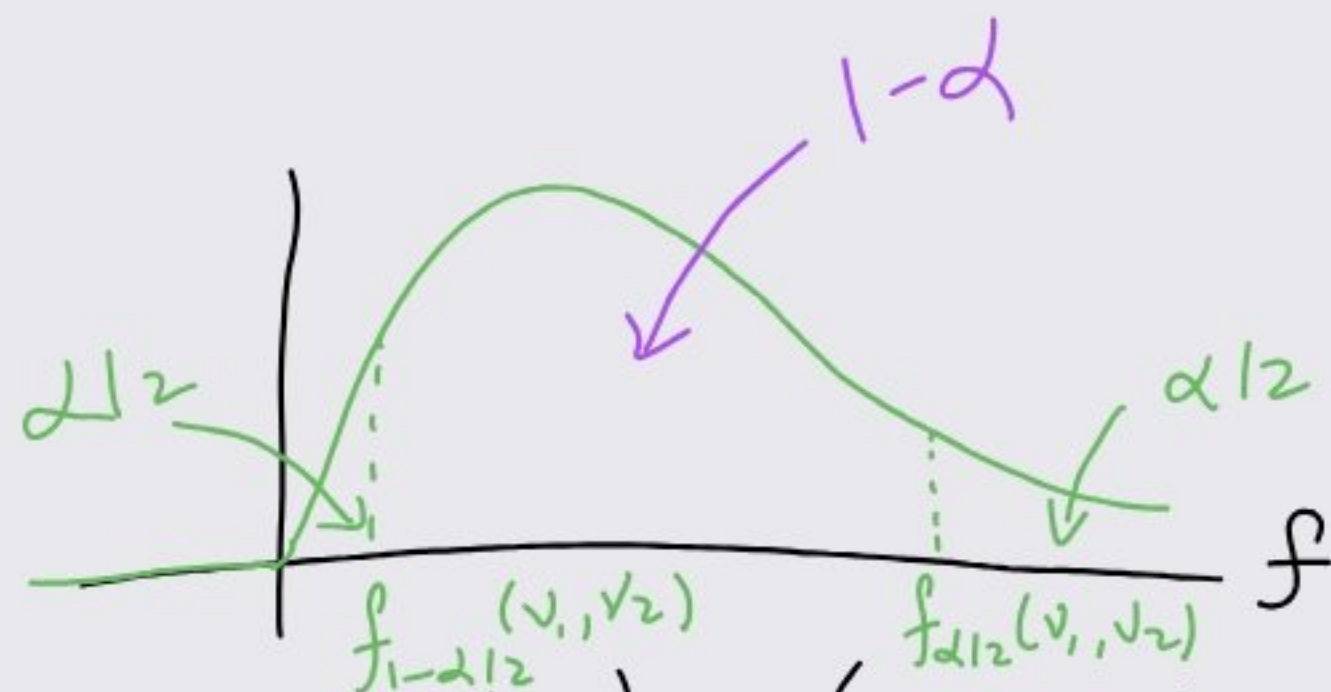
$$R = 0.135 < \sigma^2 < 0.953$$

Dos muestras: Estimación de la razón de dos varianzas.

El estadístico es:

$$F = \frac{\sigma_2^2 S_1^2}{\sigma_1^2 S_2^2}, (i)$$

que tiene una dist. F con $\nu_1 = n_1 - 1$ y $\nu_2 = n_2 - 1$ g.l. donde n_1 es el tamaño de la muestra 1 y n_2 el de la muestra 2.



El intervalo de confianza se calcula a partir de la probabilidad

$$P(f_{1-\alpha/2}(\nu_1, \nu_2) < F < f_{\alpha/2}(\nu_1, \nu_2)) = 1 - \alpha \quad (ii)$$

De (i) y (ii) se obtiene el intervalo de confianza $\rightarrow f_{1-\alpha}(\nu_1, \nu_2) = \frac{1}{f_{\alpha}(\nu_2, \nu_1)}$

$$\frac{S_1^2}{S_2^2} \cdot \frac{1}{f_{\alpha/2}(\nu_1, \nu_2)} < \frac{\sigma_1^2}{\sigma_2^2} < \frac{S_1^2}{S_2^2} \cdot f_{\alpha/2}(\nu_2, \nu_1)$$

$$\int_0^{f_{\alpha/2}(\nu_1, \nu_2)} h(x; \nu_1, \nu_2) dx = 1 - \frac{\alpha}{2}$$

$$\int_0^{f_{\alpha/2}(\nu_2, \nu_1)} h(x; \nu_2, \nu_1) dx = 1 - \frac{\alpha}{2}$$

Example 9.12: A study was conducted by the Department of Zoology at the Virginia Tech to estimate the difference in the amounts of the chemical orthophosphorus measured at two different stations on the James River. Orthophosphorus was measured in milligrams per liter. Fifteen samples were collected from station 1, and 12 samples were obtained from station 2. The 15 samples from station 1 had an average orthophosphorus content of 3.84 milligrams per liter and a standard deviation of 3.07 milligrams per liter, while the 12 samples from station 2 had an average content of 1.49 milligrams per liter and a standard deviation of 0.80 milligram per liter. Find a 95% confidence interval for the difference in the true average orthophosphorus contents at these two stations, assuming that the observations came from normal populations with different variances.

Solution: For station 1, we have $\bar{x}_1 = 3.84$, $s_1 = 3.07$, and $n_1 = 15$. For station 2, $\bar{x}_2 = 1.49$, $s_2 = 0.80$, and $n_2 = 12$. We wish to find a 95% confidence interval for $\mu_1 - \mu_2$.

where $f_{\alpha/2}(v_1, v_2)$ is an f -value with $v_1 = n_1 - 1$ and $v_2 = n_2 - 1$ degrees of freedom, leaving an area of $\alpha/2$ to the right, and $f_{\alpha/2}(v_2, v_1)$ is a similar f -value with $v_2 = n_2 - 1$ and $v_1 = n_1 - 1$ degrees of freedom.

As in Section 9.12, an approximate $100(1 - \alpha)\%$ confidence interval for σ_1/σ_2 is obtained by taking the square root of each endpoint of the interval for σ_1^2/σ_2^2 .

Example 9.19: A confidence interval for the difference in the mean orthophosphorus contents, measured in milligrams per liter, at two stations on the James River was constructed in Example 9.12 on page 290 by assuming the normal population variance to be unequal. Justify this assumption by constructing 98% confidence intervals for σ_1^2/σ_2^2 and for σ_1/σ_2 , where σ_1^2 and σ_2^2 are the variances of the populations of orthophosphorus contents at station 1 and station 2, respectively.

Solution: From Example 9.12 we have $n_1 = 15$, $n_2 = 12$, $s_1 = 3.07$, and $s_2 = 0.80$.